

NeuroNav-X

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AI/ML-based Intelligent Dead Reckoning for Seamless Navigation through GNSS-Denied Conditions

Team Learning Guide — hybrid estimator: learned IMU correction feeding an error-state EKF, constrained by vehicle kinematics and a road graph, with calibrated uncertainty. Covers tech stack, dataset quirks, every module, all formulas, baselines B0-B6, Phases 1-16, mobile app, use cases, demo script, and teacher Q&A.

How to use: read cover-to-cover once, then use the 1-page cheat-sheet (last page) for revision before judging.

1. Problem & Objectives

Problem. GNSS fails in tunnels, urban canyons, parking basements and intentional denial. A phone's raw inertial navigation (accelerometer + gyro integration) drifts tens of percent within a minute, so navigation stops exactly when it is needed most. **Goal:** keep a consumer smartphone navigating with **drift ratio <10%** (final position error / distance travelled) through 10-120 s blackouts, using **smartphone-only inputs** at deployment.

Design philosophy (README): 'Hybrid estimator, not a black box.' A tiny learned speed model corrects IMU errors; a classical 6-state error-state EKF (ES-EKF) does the fusion; vehicle kinematics and (optionally) a road graph constrain it; uncertainty is calibrated. Vehicle columns are prefixed **ref_** and used only as ground truth/supervision — enforced by tests/test_feature_contract.py.

Success criteria / Gates

Gate 0: which IO-VNBD sessions are usable (audit). Gates 1-3: baselines + learned speed reproducible end-to-end. Gate 4: map matching (FAILED by design — kept off). Gate 5: runs on a physical phone within latency budget (CLOSED on V2422 ARM64 Android 16, 0.177 ms p95 vs 100 ms budget). Open: a completed below-10% moving field test.

2. Tech Stack & Architecture

Layer	Technology	Why / where used
ML / science	Python 3.10+, PyTorch, numpy, pandas	SpeedTCN training (scripts/train_tcn.py), evaluation, plots
Model export	TorchScript, ONNX Runtime, ExecuTorch	Onnx MLPAcks; (verified 23 KB) diff 4.8e-06 vs eager
Android	Kotlin 2.1.20, Gradle 9.4.1, AGP 8.1.3, objd 25, isd 36	SpeedTCN + benchmark (mobile/)
Maps	Google Maps SDK + MapLibre Native	Offline tile map requests (MAPS_MPG_KEN in Nigeria) + local properties
Roads	OSM Overpass cache .npz 8.3 MB, ckdTee on Dijkstra	ckdTee on Dijkstra
Data	IO-VNBD via Git LFS into data/raw/	Phone V-*.csv + vehicle V-*.csv paired at 10 Hz

Data flow (read as diagram)

Phone IMU (100 Hz) + GNSS (1 Hz) → **Calibration** (gravity, forward angle, yaw channel, gyro bias) → **Features** (6 vehicle-frame signals) → **SpeedTCN** (speed + sigma, 1 Hz) → **6-state ES-EKF** (position, heading, speed, gyro bias, yaw scale) → **BlackoutManager** (AIDED/BLACKOUT/REACQUIRING) → **TrackSmoother** (display-only slew, no teleport) → Map UI

(faint estimate + bright display, orange when unaided, confidence circle).

3. Dataset IO-VNBD — 9 Quirks You Must Know

Phone logs S-*.csv + vehicle logs V-*.csv (same drive, row-for-row at true 10 Hz: position ~9.27 Hz, velocity, heading, wheel speeds, yaw-rate, longitudinal accel to ± 0.59 g). After gating: ~6.2 h reliably paired; 7 segments / 8.9 h trainable across drivers A/B/E. No phone session has BOTH good GNSS and live accel — hence pairing + simulated aiding.

#	Quirk	Handling
1	Files concatenate multiple drives (time reset, split on S-52 rows; 1866/36185) 8 km phantom path	Split on S-52 rows; 1866/36185
2	GPS SPEED (Kmh) is actually m/s	No /3.6 (MAE 1.8 vs 5.6); guarded by test
3	Phone GNSS sample-and-hold down to 0.1 Hz (median gap 0.5 s)	Simulated aiding from vehicle ref: 5 m noise, 0.5 m/s speed sigma
4	ORIENTATION cols contradict gravity (corrected)	Never used; gravity is attitude reference
5	accel-minus-gravity is high-pass filtered (brake 2.0 Hz, rate 0.18)	Can't distinguish
6	Gyro yaw channel differs per family (gy on S-52)	Selected from data (select_yaw_channel); Phase 13 uses gravity-projected yaw
7	S-A4 stray empty column shifts sensors	Header-text matching + drop all-NaN cols
8	V-*.csv is dense truth	All ref_ prefixed; supervision/eval only
9	Phone/vehicle offsets up to ~9 s (one +115 s)	2 s-smoothed cross-correlation, gate corr $\geq$ 0.35; shift(-lag); paired 8→14 sessions

Simulated aiding (src/neuronav/data/gnss_sim.py): subsample truth to 1 Hz, 5 m horizontal noise, accuracy 5 m, bearing Doppler-style 3 deg withheld below 2 m/s. Differencing noisy fixes would give ~40-70 deg heading error — hence Doppler bearing. Blackout windows require mean speed $\geq$ 5 m/s, distance $\geq$ 20 m.

4. Code Tour (what lives where)

src/neuronav/data/

loader.py — 18+6 columns, PRODUCTION_FEATURES=[ax,ay,az,gx,gy,gz,grav_x,grav_y,grav_z], mag optional/excluded. **reference.py** — SI scales (km→m $\times$ 1000, km/h→m/s /3.6, g→9.80665), estimate_sync_lag coarse-to-fine. **gnss_sim.py** — aiding + apply_blackout + pick_blackout_windows. **own_drive.py** — own-drive CSV schema + gap splits + summarise.

src/neuronav/calibration/

signals.py — gravity_unit, angular_rate, heading_rate=-sum(omega.ghat) (clockwise positive), horizontal basis e1/e2, tilt. **alignment.py** — Alignment dataclass, robust_line (MAD trim), select_yaw_channel, estimate_gyro_bias, estimate_alignment (joint regress h1,h2 vs dv/dt → angle=atan2).

src/neuronav/fusion/

es_ekf.py — PlanarESEKF (see formulas). **blackout.py** — BlackoutManager + TrackSmoother. **run.py** — run_es_ekf / run_raw_ins / constant_velocity_baseline, SPEED_MODE_ACCEL/HOLD/TCN.

src/neuronav/models/

dataset.py — 6 features, 6 s windows stride 0.5 s causal label, FeatureScaler (train-only fit). **tcn.py** — CausalConv1d, TemporalBlock, SpeedTCN, gaussian_nll. **predictor.py** — batched 4096 sliding windows, affine invert, sigma scaling.

src/neuronav/mapmatch/ + evaluation + scripts + mobile

mapmatch: osm.py fetch/cache, graph.py KD-tree + Dijkstra, hmm.py offline Viterbi, online.py causal multi-hypothesis, apply.py 2 Hz post-hoc. **evaluation/metrics.py:** path_length, drift_ratio, ate_rmse, final_position_error. **scripts/:** download_data, audit_paired (Gate 0), train_tcn (Gate 2, --held-out B), run_baselines (Gates 1/3), export_model, profile_edge, eval_recovery, demo_replay (finale fallback), analyze_own_drive / analyze_live_blackout. **mobile/:** nav/*.kt ports (Signals, Calibration, Features,

SpeedModel, EsEkf, Blackout, Navigator, LiveSource/ReplaySource, HeadingRateAccumulator, LiveSpeedAdapter, StationaryHold, FieldBlackoutTest), dual-track UI, FullNavigationLoopBenchmark.

5. Math & Formulas (the heart of the project)

5.1 SpeedTCN

Causal conv pads LEFT only: $y[t]$ depends on $\leq t$. Block: $h = \text{Drop}(\text{GELU}(\text{GN}(\text{Conv1}(x))))$, $h = \text{Drop}(\text{GELU}(\text{GN}(\text{Conv2}(h))))$, $\text{out} = \text{GELU}(h + \text{Res}(x))$ with $\text{GroupNorm}(1, C)$. SpeedTCN: 4 levels, 32 channels, $k=3$, dropout 0.1, dilations 1,2,4,8. Head: $\text{Linear}(C, C) \rightarrow \text{GELU} \rightarrow \text{Linear}(C, 2)$; $\text{mean} = \text{softplus}(\text{out}_0) \geq 0$, $\text{logvar} = \text{clamp}(\text{out}_1, -6, 6)$. Receptive field $\text{RF} = 1 + 2 \cdot (k-1) \cdot (2^L - 1) = 1 + 2 \cdot 2 \cdot 15 = \mathbf{61 \text{ samples} = 6.1 \text{ s @ 10 Hz}}$, 24,194 params. Loss Gaussian NLL: $\mathbf{0.5 \cdot (\exp(-\text{logvar}) \cdot (y - \text{mean})^2 + \text{logvar})}$. Features: $[a_{\text{forward}}, a_{\text{lateral}}, a_{\text{vertical}} = \text{sum}(a_{\text{up}}) - 9.80665, \text{yaw_rate}, a_{\text{horiz_mag}}, \text{jerk}]$. Skill = $\mathbf{1 - \text{RMSE} / \text{RMSE_mean}}$: B 1-3.23/5.92=0.455, A 1-3.05/5.75=0.470, corr ~0.83. Fused at ~1 Hz (10 Hz double-counts the correlated 6 s window).

5.2 Six-state ES-EKF

Nominal $\mathbf{x} = [pE, pN, \text{psi}, v, b_w, s_w]$ (east, north, heading clockwise from north, speed, gyro bias, yaw-scale error). Error dx with covariance P , inject-then-reset, $\text{wrap}(a) = (a + \pi) \% 2\pi - \pi$. **Propagate**: $w_{\text{deb}} = \text{yaw_meas} - b_w$; $w = (1 + s_w) \cdot w_{\text{deb}}$; $pE += v \cdot \sin(\text{psi}) \cdot dt + 0.5 \cdot a \cdot \sin(\text{psi}) \cdot dt^2$; $pN += v \cdot \cos(\text{psi}) \cdot dt + 0.5 \cdot a \cdot \cos(\text{psi}) \cdot dt^2$; $\text{psi} = \text{wrap}(\text{psi} + w \cdot dt)$; $v += a \cdot dt$. Jacobian F : $F[pE, \text{psi}] = v \cdot \cos \cdot dt$, $F[pE, v] = \sin \cdot dt$, $F[pN, \text{psi}] = -v \cdot \sin \cdot dt$, $F[pN, v] = \cos \cdot dt$, $F[\text{psi}, b] = -(1 + s_w) \cdot dt$, $F[\text{psi}, s] = w_{\text{deb}} \cdot dt$. Q : pos $(0.5 \cdot 0.6 \cdot dt^2)^2$, heading $(0.02^2 + (0.01 \cdot |w|)^2) \cdot dt$, speed $(0.6 \cdot dt)^2 + (0.4^2) \cdot dt$, bias $(2e-4)^2 \cdot dt$, scale $(1e-5)^2 \cdot dt$. **Update** $_apply(H, y, R)$: $S = HPH' + R$, $NIS = y'S - y$ rejected beyond 5σ (streak inflates P for re-acquire), $K = PH'S^{-1}$, Joseph form $P = (I - KH)P(I - KH)' + KRK'$. Measurements: position $\sigma = \max(\text{acc}, 3.0)$; speed $\sigma = 0.7$; Doppler bearing $\sigma = \text{bearing_acc}$ else 0.15 gated speed $\geq 2 \text{ m/s}$ (5 native); learned speed $\sigma = \max(\text{net_sigma}, 0.8)$ only when dark+valid at 1 Hz — never writes position/heading directly (graceful); map anisotropic $R = \text{Rot} \cdot \text{diag}(\text{along}^2, \text{cross}^2) \cdot \text{Rot}'$; ZUPT $y = 0 - v$ $\sigma = 0.05$ below 0.3 m/s.

5.3 Metrics / calibration / map / blackout

Metrics: $\text{path_length} = \sum ||\text{ref}[i] - \text{ref}[i-1]||$; $\text{drift} = ||\text{est_end} - \text{ref_end}|| / \text{path_length}$ (target <10%); $\text{ATE} = \sqrt{\text{mean} ||\text{est} - \text{ref}||^2}$; $\text{FPE} = ||\text{est_end} - \text{ref_end}||$. **Calibration**: $\text{ghat} = g / ||g||$; $\text{heading_rate} = -\omega \cdot \text{ghat}$; forward angle = $\text{atan2}(c1, c0)$ from $\text{lstsq}([h1, h2], dv/dt)$; yaw channel = max corr of $\int \text{gyro}$ vs d_{bearing} . **Map HMM**: emission $-0.5 \cdot (d/s)^2 + \text{heading} \cdot w \cdot (1 - \cos)$, transition $-|\text{route-travelled}|/12$, Viterbi offline; online multi-hypothesis needs $\text{conf} \geq 0.60$ for 4 steps and $\sigma \leq 18 \text{ m}$. **Blackout**: loss if gap >3 s or $\text{acc} > 30 \text{ m}$; reacquire 3 consistent fixes in 6 s within 40 m; display slew rate = $\text{clip}(\text{dist}/2, 40, 200) \text{ m/s}$ — estimate untouched (snap is Bayes-optimal).

6. Baselines B0-B6 — Results on Held-Out Driver B

Median drift (p90 in brackets) vs 10 Hz vehicle truth. B0 const-velocity floor; B1 raw INS open-loop; B3 ES-EKF classical; **B5 TCN+ES-EKF shipped**; B4/B4b map off unless --map; B6 adds yaw-scale (deployment_config enables scale only).

Blackout	B0 const-vel	B1 raw INS	B3 ES-EKF	B5 TCN+ES-EKF
10 s	15.7%	14.8%	9.7% (55.7)	10.5% (34.7)
30 s	38.5%	10.0%	10.8% (56.1)	16.3% (32.0)
60 s	69.2%	17.9%	15.0% (61.1)	14.0% (34.6)
120 s	89.1%	28.9%	26.0% (91.3)	14.7% (51.2)

Reading: learned speed **halves median drift at 120 s and cuts p90 35-45% at every duration**. Speed model RMSE 3.2 vs 5.9 naive (skill 0.46), 24k params. Oracle ablations: on B5, true speed cuts 30/60/120 s 16.3→5.6, 14.0→8.5, 14.7→10.8; both speed+heading $\leq 0.8\%$ (mechanization sound — remaining error is speed). Map Gate 4 FAILS: needs error < ~17 m road spacing; beyond that confident wrong-road match drags solution (naive online 60 s 27.4→39.7%); gated still loses on B. Recovery jump drops 91-95% (120 s

173.7→8.8 m) with accuracy 10 s later unchanged ± 0.3 m.

7. Phases 1-16 (one line each)

1 Screening: classical ladder + 300 s warmup. 2 Learned speed (B5): 24k TCN, 1 Hz fusion, halves 120 s drift. 3 Map: HMM+online, Gate 4 fails → display-snapping only. 4 Edge: TorchScript/ONNX/XNNPACK verified, desktop 0.74 ms (0.7% of 100 ms), INT8 rejected (+7.8% error). 5 Blackout machine + demo_replay.py fallback (no GNSS/vehicle/internet). 6 Physics B6: yaw-scale state (bias $\propto$ time, scale $\propto$ turn; M_seg00 49.9°=4.4+44.1); horizon/balance off. 7 Kotlin port: parity <0.1 m after fixing bearing_acc (171 m lesson) + decimation. 8 Real-device: V2422 full-loop 0.024/0.177 ms → Gate 5 closed; field protocol. 9 Own-drive evidence pipeline. 10 Retrain with representative validation REGRESSED → deliberately not deployed. 11 One-button raw+diagnostics+summary recorder. 12 Controlled 60 s estimator-only withholding + calibration gate. 13 Native full-rate gravity-projected yaw (fixes 51.5% field blowup → 7.2% counterfactual). 14 Crash/bias/scale fixes; live TCN +4.5-5 m/s bias found. 15 LiveSpeedAdapter learns additive bias from aided pairs. 16 Motion-aware calibration + stationary hold + gap reset (12:04 replay 55.7→25.0%); **below-10% completed moving field test still open.**

8. Mobile App & Edge

Logger schema: timestamp_ms,ax,ay,az,gx,gy,gz,mx,my,mz,grav_x,grav_y,grav_z,lat,lon,alt,speed,accuracy,bearing,bearing_acc,gps_fresh (monotonic elapsedRealtimeNanos; gps_fresh disambiguates hold; Doppler bearing not differenced). Live Navigator: 100 Hz callbacks → integer decimation → CALIBRATING (200 angles+100 baselines, ~350 s demo) → 6-feet ring → TCN 1 Hz → ES-EKF → manager → smoother. Arm 60 s test: parked → 15 s lead-in → TEST WITHHOLDING (raw kept as hidden ref) → ordinary recovery. UI: Google/MapLibre GN-offline/PMTiles/canvas fallback; faint estimate + bright display (orange unaided); confidence circle; status shows 1σ /distance never 'drift' (no truth live). Benchmark screen: bundled drive 5× through callbacks/cal/feats/TCN/EKF/manager/smoother; run 2× cool+hot. Artifacts per session: nav_drive_100Hz.csv + nav_diagnostics_10Hz.csv + nav_summary.json.

9. Use Cases & Demo Script

Use cases: tunnels/urban canyons/parking/denial on consumer phones; fleet continuity; offline Greater Noida maps; field data collection (unfiltered accel + 1 Hz GNSS); teaching hybrid-vs-black-box and audit-first ML. **Demo (5 min):** (1) show aided stable + status; (2) trigger blackout — overlay truth vs raw INS (diverges) vs NeuroNav-X (holds), orange unaided + growing sigma; (3) recovery — estimate snaps, display slews in ~2-4 s, no teleport, sigma collapses; (4) fallback: demo_replay.py replays real sensor CSV through the real estimator with no GNSS/vehicle/connectivity (B 120 s/1236 m: 24.6% vs 48.8% raw, jump 15.5 m, heading 16.6°).

10. Teacher / Judge Q&A; (memorize these)

Question	Model answer (3-4 lines)
Why not integrate accel?	Phone accel is high-pass filtered (brake -2.5 shows -0.18, corr 0.02). TCN learns speed from vibration/motor cues
Why speed, not position?	Oracle: true speed fixes most error at all durations; network never writes pos/heading, so failures degrade gracefully
Why fuse at 1 Hz?	6 s window is correlated; 10 Hz double-counts and lost short outages. 1 Hz fixed it.
Why does map fail?	Needs error < ~17 m road spacing; beyond, confident wrong-road match drags solution + sigma miscalibrated 0.5
Why smooth display, not filter?	Snap is Bayes-optimal; inflating R for cosmetics worsens the estimate. Smoother is display-only.
Why yaw-scale state?	Heading error = bias·T + scaleErr·turn; M_seg00 44/50 deg is scale. Helps B p90 -6-7 pts.
Why not INT8 / Transformer?	fp32 already 0.7% of budget; INT8 costs 7.8% accuracy. Transformer stays desktop until Android export+latency
Biggest field lessons?	bearing_acc missing (171 m), 17% decimation drop, double gyro bias, frozen gz, live TCN +5 bias, 1479 s gap →
What is still open?	A completed below-10% moving field test; disturbed stops + difficult sessions remain.

11. Key Numbers, 2-Minute Pitch, Cheat-Sheet

Metric	Value
Speed RMSE	3.2 m/s vs 5.9 naive (skill 0.46), corr 0.83, 24k params, RF 6.1 s
Drift B5 (med/p90)	10 s 10.5/34.7, 30 s 16.3/32.0, 60 s 14.0/34.6, 120 s 14.7/51.2%
Recovery	Jump -91-95% (120 s 173.7→8.8 m); +10 s accuracy ±0.3 m
Latency	Desktop 0.74 ms; V2422 full-loop 0.024 med / 0.177 p95 ms (0.18% of 100 ms)
Parity	Desktop vs Android <0.1 m on 3 routes × 2 drivers
Data	14 paired (12.1 h), 7 trainable (8.9 h), drivers A/B/E
Field	First controlled 19.2%, regressed 114% → fixed; 3 Sep heading 2.9-4.8 deg; Phase16 replay 55.7→25.0%

2-minute pitch

"GNSS dies in tunnels and urban canyons. Phone inertial navigation drifts away in a minute because the accelerometer is filtered and the gyro has bias and scale errors. NeuroNav-X is a hybrid: a tiny 24k-parameter causal TCN learns speed from phone vibration with calibrated uncertainty, and a 6-state error-state EKF fuses it with Doppler bearing — never writing position directly so failures degrade gracefully. On a held-out driver it halves 120-second drift from 26% to 14.7% and cuts the worst cases 35-45%; recovery jumps shrink 95% with no teleport thanks to a display-only smoother. It runs in 0.18 milliseconds on a real phone and works fully offline. Map matching was tried and deliberately turned off because it fails beyond 17-metre road spacing — that negative result is documented. What remains is closing the sub-10% moving field test."

Commands

```
./envs/bin/python scripts/download_data.py # fetch      ./envs/bin/python
scripts/audit_paired.py # Gate 0      ./envs/bin/python scripts/train_tcn.py --held-out B #
Gate 2      ./envs/bin/python scripts/run_baselines.py --drivers B --checkpoint
outputs/checkpoints/speed_tcn_holdout_B.pt # Gates 1/3      ./envs/bin/python -m pytest tests/
# guards
```

Glossary

GNSS= satellite positioning; INS= inertial navigation; dead reckoning= integrate motion from last fix; ES-EKF= error-state extended Kalman filter (estimates errors, inject-then-reset); TCN= temporal convolutional network (causal dilated conv); HMM= hidden Markov model (map matching); ATE= absolute trajectory error; Doppler bearing= velocity-derived heading; ZUPT= zero-velocity update; NIS= normalized innovation squared (outlier gate); LODO= leave-one-driver-out.